A VOF-Masked PINN Surrogate for Rapid Flow Prediction in an Argon-Stirred Ladle: Cross-Flow-Rate and Cross-Position Generalization with Multi-Baseline Benchmarking

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Abstract

Repeated computational fluid dynamics (CFD) simulations of argon-stirred ladle flow are computationally prohibitive when multiple operating conditions must be evaluated. This study develops a volume-of-fluid (VOF)-masked physics-informed neural network (PINN) surrogate for rapid three-dimensional prediction of the statistically steady liquid-steel flow field under varying gas flow rates (Q) and injection positions (m). A three-dimensional VOF-based CFD model generates reference datasets for five center-blowing conditions (Q = 40–120 NL/min) and three eccentric-blowing conditions. The training framework employs soft VOF-based pointwise weighting to restrict physics constraints to the steel phase, eliminating explicit interface boundary conditions while maintaining mathematical rigor. An 8×384 SiLU-activated multilayer perceptron achieves $R^2 = 0.983$ for center-blowing interpolation and $R^2 = 0.986$ for extrapolation, with no perceptible degradation beyond the training range. A six-baseline comparison establishes the performance hierarchy: the proposed model outperforms DeepONet ($R^2 = -0.72$), while Kriging and POD-ROM face scalability limitations on the 9.3×10^5-cell unstructured mesh. Cross-position generalization is enabled through a binary injection-position parameter $m \in \{0, 1\}$; joint training recovers $R^2 = 0.94$–0.97 for eccentric-blowing predictions, whereas a

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center-only model collapses ($R^2 < 0$). Multi-seed ensemble analysis confirms stability (0.972 ± 0.003), and a continuity-only physics constraint improves R^2 by +0.014 over purely data-driven training. The surrogate reduces per-case prediction time from approximately 36 hours to 5 seconds, providing a practical tool for operating-window exploration in ladle-furnace refining.

Keywords: Computational fluid dynamics; Physics-informed neural network; Surrogate modeling; Ladle furnace; Argon-stirred flow; Volume-of-fluid; Cross-position generalization

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1. Introduction

Ladle furnace (LF) refining is a critical secondary metallurgy process for controlling steel cleanliness, thermal uniformity, and compositional homogeneity. During bottom argon stirring, the injected gas forms a buoyancy-driven plume that induces strong recirculation in the molten steel bath. This plume-driven motion enhances mixing and promotes the upward transport of inclusions toward the slag layer [1, 2]. At the same time, excessively strong stirring may intensify slag–metal interfacial deformation, expose the steel to the atmosphere (slag eye formation), and increase the risk of re-oxidation and slag entrainment [3]. Reliable prediction of the internal flow field is therefore fundamental to process analysis and optimization.

Computational fluid dynamics (CFD) has become a standard tool for studying ladle hydrodynamics. Euler–Euler, volume-of-fluid (VOF), and hybrid multiphase models have been used to investigate gas–liquid interaction, plume evolution, interfacial deformation, and inclusion transport [4, 5]. These studies have substantially improved the understanding of ladle flow behavior. However, high-fidelity three-dimensional VOF simulations remain computationally expensive: a single steady-state case may require upwards of 36 hours. When multiple gas flow rates and injection configurations must be evaluated—a common requirement in industrial practice—repeated CFD becomes prohibitive.

In recent years, machine learning methods have shown growing potential for surrogate modeling in fluid mechanics. Once trained, neural networks can provide predictions orders of magnitude faster than repeated numerical simulation [6]. However, purely data-driven models depend heavily on the amount and coverage of labeled data, which is a practical difficulty because training data are typically obtained from expensive CFD. Physics-informed neural networks (PINNs) address this limitation by incorporating governing-equation residuals and boundary information into the training process [7, 8]. PINNs have been applied to various fluid-mechanics problems [9], but their application to industrial ladle-flow prediction remains limited.

A key challenge for PINN-based ladle-flow surrogates is the multiphase nature of the system. The ladle contains three immiscible phases (argon gas, molten slag, and liquid steel) whose densities span four orders of magnitude (~ 1, ~ 3000, and ~ 7000 kg/m^3). Enforcing the Navier–Stokes residuals across the entire three-phase domain is numerically ill-conditioned because the momentum magnitude varies by orders of magnitude between phases. Previous PINN studies on multiphase flow have either simplified the phase physics or required extensive interface-tracking data [10, 11].

In this study, we propose a VOF-masked PINN framework that circumvents the multiphase challenge through domain decomposition based on the steel volume fraction α_{steel}. The key insight is that in pure-steel cells ($\alpha_{\text{steel}} \to 1$), the VOF mixture equations reduce rigorously to single-phase incompressible RANS. By restricting the physics constraints to the steel phase through soft VOF-based pointwise weighting, the surrogate avoids explicit interface boundary conditions while retaining mathematical consistency where it matters.

The contributions of this work are:

1. A VOF-masked PINN framework that restricts physics constraints to the steel phase through soft pointwise weighting, eliminating explicit interface boundary conditions.
2. Binary injection-position parameterization ($m \in \{0, 1\}$) enabling a single surrogate to predict both center-bottom and eccentric-bottom blowing

configurations.

3. Six-baseline comparison (DeepONet, Kriging, POD-ROM, KAN, and two MLP variants) establishing a performance benchmark for the ladle-flow surrogate problem.
4. Uncertainty quantification via multi-seed ensemble and Monte Carlo dropout.
5. A practical surrogate achieving $R^2 = 0.983$ for center-blowing and $R^2 = 0.94$–0.97 for eccentric-blowing, with near-zero extrapolation degradation and 5-second inference.

2. CFD Model

2.1. Geometry and Computational Domain

The ladle geometry was modeled as a truncated cone representative of an industrial 100-ton LF vessel. The computational domain encompasses the entire internal fluid volume, including the molten steel, slag layer, and upper gas space. The main dimensions are summarized in Table 1. Argon is introduced through a bottom nozzle located at the center of the ladle base (center-blowing, $m = 0$) or at an eccentric position at one-third radius (eccentric-blowing, $m = 1$).

Table 1: Main dimensions of the ladle model.

Parameter	Value
Upper diameter	2.077 m
Lower diameter	1.832 m
Bath height (steel + slag)	1.85 m
Slag layer thickness	0.15 m
Gas space height	0.69 m
Nozzle diameter	0.08 m
Center nozzle position	(0, 0, 0)
Eccentric nozzle position	(0.47, 0, 0)

2.2. Mesh Generation and Independence

An unstructured tetrahedral mesh was generated with local refinement around the gas inlet and in the central plume region. Grid-independence was assessed using three mesh levels (Table 2). The maximum plume-region velocity and volume-averaged velocity served as comparison indices. The medium mesh with 9.32×10^5 cells was adopted, as further refinement to 1.36×10^6 cells changed the maximum velocity by only 1.14%.

Table 2: Mesh-independence study.

Level	Cells	$\|U\|_{\max}$ (m/s)	Rel. diff. (%)
Coarse	612,450	0.392	10.09
Medium	932,080	0.431	1.40
Fine	1,356,740	0.436	—

2.3. Governing Equations and Turbulence Modeling

The multiphase flow is described using the VOF formulation with three immiscible phases: argon (gas), molten slag, and liquid steel. The mixture continuity and momentum equations are:

$$\nabla \cdot \mathbf{u} = 0 \tag{1}$$

$$\rho(\boldsymbol{\alpha}) \frac{D\mathbf{u}}{Dt} = -\nabla p + \nabla \cdot \left[\mu(\boldsymbol{\alpha}) \left(\nabla \mathbf{u} + \nabla \mathbf{u}^T\right)\right] + \mathbf{F}_\sigma \tag{2}$$

where $\boldsymbol{\alpha} = (\alpha_{\text{steel}}, \alpha_{\text{slag}}, \alpha_{\text{argon}})$ denotes the phase volume fractions ($\sum \alpha_i = 1$), and $\rho(\boldsymbol{\alpha}) = \sum \alpha_i \rho_i$, $\mu(\boldsymbol{\alpha}) = \sum \alpha_i \mu_i$ are mixture properties. The term $\mathbf{F}_\sigma$ represents surface tension at phase interfaces.

Turbulence is modeled using the realizable k–ε model with standard wall functions. The choice of the realizable variant is motivated by its known superiority in predicting axisymmetric jet and plume flows compared to the standard k–ε model [12]. The discrete phase model (DPM) is used for argon bubble injection, allowing the bubbles to exchange momentum with the continuous phase through drag forces without being resolved in the VOF field—a standard approach in ladle-flow CFD [13].

2.4. Boundary Conditions and Numerical Settings

The bottom wall and side walls are modeled as no-slip boundaries. The top surface of the gas space is treated as a pressure outlet. Argon is injected through the bottom nozzle as DPM particles with a velocity prescribed by the gas flow rate. All simulations were carried out in ANSYS Fluent using the SIMPLE pressure–velocity coupling and second-order spatial discretization. Each case was run to a statistically steady state ($t = 300$ s for steady-state export).

2.5. Dataset Generation and Quality Verification

The CFD solution was exported at cell centers in CSV format with 12 columns:

```
cellnumber, x-coordinate, y-coordinate, z-coordinate, pressure,
x-velocity, y-velocity, z-velocity, argon-vof, steel-vof,
slag-vof, air-vof
```

where `air-vof` is a residual Fluent column (effectively zero in all cases). Five center-blowing conditions ($Q = 40, 60, 80, 100, 120$ NL/min) and three eccentric-blowing conditions ($Q = 40, 80, 120$ NL/min) were generated, each containing 932,079 cells.

Stringent data-quality checks were applied. All eight datasets satisfy $\sum \alpha_i \equiv 1$ for every cell (100% verification). The vertical phase stratification is consistent across flow rates: steel dominates below $z \approx 1.85$ m, slag occupies the intermediate $z \approx 1.85$–2.05 m layer, and argon fills the upper space above $z \approx 2.05$ m. Velocity monotonicity with Q was verified for all 18 vertical layers: the mean velocity magnitude $|\mathbf{U}|$ increases strictly with Q at every level (zero violations), as shown in Table 3.

Table 3: Velocity monotonicity verification: mean $|\mathbf{U}|$ (m/s) at selected heights.

z (m)	$Q = 40$	$Q = 60$	$Q = 80$	$Q = 100$	$Q = 120$
0.5	0.009	0.010	0.011	0.012	0.014
1.0	0.017	0.022	0.025	0.027	0.031
1.5	0.027	0.033	0.037	0.039	0.043
1.8 (interface)	0.044	0.053	0.059	0.063	0.069

2.6. CFD Validation Justification

The CFD model serves as the reference "teacher" for surrogate training, a paradigm established in CFD-informed surrogate modeling [14]. The consistency of eight operating-condition datasets—identical mesh topology, uniform phase stratification, monotonic velocity scaling, and perfect VOF sum—provides internal validation. While comparison with experimental measurements would strengthen the reference, the present study prioritizes demonstration of surrogate prediction against a numerically consistent CFD baseline, which is the standard approach for computational PINN surrogates [7]. Experimental validation of the CFD model itself is deferred to future work.

3. Methodology

3.1. Overall Framework

The proposed CFD–VOF–PINN framework comprises four stages (Fig. 1): (1) CFD data generation under multiple operating conditions, (2) VOF-informed data preprocessing and domain decomposition, (3) neural network training with physics constraints restricted to the steel phase, and (4) surrogate deployment for rapid multi-condition prediction.

3.2. VOF-Informed Domain Decomposition

The central methodological contribution is the treatment of the multiphase domain. In a three-phase VOF system, the mixture velocity $\mathbf{u}$ is shared across phases. In pure-steel cells ($\alpha_{\text{steel}} \to 1$):

$$\rho(\boldsymbol{\alpha}) \to \rho_{\text{steel}}, \quad \mu(\boldsymbol{\alpha}) \to \mu_{\text{steel}}, \quad \mathbf{F}_\sigma \to 0 \tag{3}$$

Under these conditions, the VOF mixture equations (Eqs. 1, 2) reduce rigorously to the single-phase incompressible RANS equations. This mathematical property—not computational limitation—motivates restricting physics constraints to the steel region.

A soft VOF-based pointwise weight is introduced:

[Schematic of the CFD–VOF–PINN framework — to be inserted]
(1) CFD data generation → (2) VOF filtering → (3) PINN training → (4) Surrogate prediction

Figure 1: Schematic of the proposed CFD–VOF–PINN surrogate framework.

$$w_{\text{data}}(\mathbf{x}) = \text{clamp}(\alpha_{\text{steel}}(\mathbf{x}),\ 0.01,\ 1.0) \tag{4}$$

$$w_{\text{pde}}(\mathbf{x}) = w_{\text{data}}(\mathbf{x}) \cdot \frac{1}{2}\left[1 + \cos\left(\pi \cdot \min\left(\frac{z_{\text{top}} - z}{T}, 1\right)\right)\right] \tag{5}$$

where $z_{\text{top}} = 1.85$ m is the steel–slag interface height and $T = 0.20$ m is the transition length scale. The spatial decay $w_{\text{pde}}(z)$ smoothly deactivates physics constraints as the interface is approached, eliminating the need for a hard interface boundary condition. Table 4 documents the verified phase stratification that justifies this decomposition.

Table 4: Verified vertical phase stratification (mean VOF per layer, averaged across five center-blowing cases).

z range (m)	$\bar{\alpha}_{\text{steel}}$	$\bar{\alpha}_{\text{slag}}$	$\bar{\alpha}_{\text{argon}}$
0 – 1.6	1.000	0.000	0.000
1.6 – 1.85	0.731	0.269	0.000
1.85 – 2.05	0.000	0.701	0.299
2.05 – 2.55	0.000	0.000	1.000

3.3. Network Architecture

The surrogate model maps operating conditions and spatial coordinates to local flow variables:

$$\mathbf{x} = (x, y, z, Q, m) \rightarrow \hat{\mathbf{y}} = (\hat{u}, \hat{v}, \hat{w}, \hat{p}) \tag{6}$$

where Q is the argon flow rate (NL/min) and $m \in \{0, 1\}$ is the injection position (0 = center-bottom, 1 = eccentric-bottom). The network architecture is an 8×384 fully connected multilayer perceptron (MLP) with SiLU (Swish) activation:

$$\text{Input}(5) \rightarrow 384 \rightarrow 384 \rightarrow 384 \rightarrow 384 \rightarrow 384 \rightarrow 384 \rightarrow 384 \rightarrow 384 \rightarrow \text{Output}(4) \tag{7}$$

totaling 1,038,724 trainable parameters. A smaller 6×128 variant (83,844 parameters) is used for ablation studies to reduce computational cost.

Before training, all variables are normalized. Spatial coordinates are linearly mapped to $[-1, 1]$ using domain bounds. The flow rate is normalized as $Q_{\text{norm}} = 2Q/Q_{\max} - 1$. The output variables are standardized per-channel using z-score normalization:

$$\tilde{u} = (u - \mu_u)/\sigma_u, \quad \tilde{v} = (v - \mu_v)/\sigma_v, \quad \tilde{w} = (w - \mu_w)/\sigma_w, \quad \tilde{p} = (p - \mu_p)/\sigma_p \tag{8}$$

where μ_*, σ_* are computed from the steel-phase training data. Z-score normalization is critical for pressure: the hydrostatic component dominates ($p \sim 1.3 \times 10^5$ Pa), and max-absolute scaling collapses the pressure variance to near zero.

3.4. Physics Constraints

Physics constraints are incorporated as regularization terms applied within the steel-phase domain. The continuity residual is evaluated in physical space:

$$\mathcal{R}_{\text{cont}} = \frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} + \frac{\partial w}{\partial z} \tag{9}$$

Spatial derivatives are computed via automatic differentiation on the normalized network output, then scaled to physical units using the normalization parameters and domain dimensions:

$$\left.\frac{\partial u}{\partial x}\right|_{\text{phys}} = \frac{\partial \tilde{u}}{\partial x_{\text{norm}}} \cdot \frac{2\sigma_u}{L_x} \tag{10}$$

The momentum residual with mixing-length eddy viscosity is:

$$\mathcal{R}_u = u\frac{\partial u}{\partial x} + v\frac{\partial u}{\partial y} + w\frac{\partial u}{\partial z} + \frac{1}{\rho}\frac{\partial p}{\partial x} - \nu_{\text{eff}}\nabla^2 u \tag{11}$$

where $\nu_{\text{eff}} = \nu + \nu_t$, and $\nu_t = l_m^2\sqrt{2S_{ij}S_{ij}}$ with the mixing length $l_m =$ 0.01 m calibrated from CFD. The full momentum formulation is evaluated in the ablation studies (Section 5.3); the best-performing model uses only the continuity constraint.

3.5. Boundary Conditions

For the DPM-based argon injection, the continuous phase has no mass-flow inlet—the bottom nozzle face is a no-slip wall for the liquid steel. Accordingly, only wall boundary conditions are enforced:

$$\mathcal{L}_{\text{bc}} = \frac{1}{N_{\text{wall}}}\sum_{k=1}^{N_{\text{wall}}} \|\hat{\mathbf{u}}(\mathbf{x}_k)\|^2 \tag{12}$$

Wall points are filtered by $\alpha_{\text{steel}} > 0.01$ to exclude those in the slag and gas layers, yielding approximately 41,800 boundary points per case. No inlet or outlet boundary constraints are used.

3.6. Loss Function and Training

The total loss function combines data fidelity, physics regularization, and boundary enforcement:

$$\mathcal{L} = \mathcal{L}_{\text{data}} + \lambda_{\text{phys}}\mathcal{L}_{\text{phys}} + \lambda_{\text{bc}}\mathcal{L}_{\text{bc}} \tag{13}$$

where $\mathcal{L}_{\text{data}} = \frac{1}{N}\sum_i \|\hat{\mathbf{u}}_i - \mathbf{u}_i\|^2$ is the mean squared error on velocity components. Weight selection is informed by a systematic sweep (Section 5.1): $\lambda_{\text{bc}} = 0$ and $\lambda_{\text{phys}} \in [10^{-5}, 10^{-1}]$ for continuity-only training.

Training uses the Adam optimizer with an initial learning rate of 10^{-3}, a batch size of 8192, and cosine annealing to 10^{-6}. A two-phase strategy is employed: Phase 1 (500 epochs) uses data loss only, allowing the network to establish a data-driven baseline; Phase 2 activates physics constraints with a linear ramp over 100 epochs. Training terminates at 2000–3000 epochs, requiring approximately 5 minutes on an NVIDIA RTX 5070 Laptop GPU (8 GB). Once trained, the surrogate predicts the full three-dimensional flow field (9.3×10^5 cells) in approximately 5 seconds.

4. Experimental Setup

4.1. Operating Conditions

Eight CFD cases were generated for this study (Table 5). Center-blowing cases $Q = 40$, 60, and 80 NL/min form the training set (n = 2,796,237 cells after VOF filtering). Center-blowing $Q = 100$ and 120 NL/min are reserved for extrapolation testing. Three eccentric-blowing cases ($Q = 40$, 80, 120 NL/min) are used for cross-position generalization evaluation.

Table 5: Operating conditions and dataset partitioning.

Q (NL/min)	Position	m	Cells	Role
40	Center	0	932,079	Training
60	Center	0	932,079	Training
80	Center	0	932,079	Training
100	Center	0	932,079	Extrapolation test
120	Center	0	932,079	Extrapolation test
40	Eccentric	1	932,079	Cross-position test
80	Eccentric	1	932,079	Cross-position test
120	Eccentric	1	932,079	Cross-position test

4.2. Baseline Methods

Six methods are compared to establish the performance landscape for the ladle-flow surrogate problem:

1. **MLP** (6×128): Same architecture as the PINN backbone, no physics constraints ($\lambda_{\text{phys}} = \lambda_{\text{bc}} = 0$).

2. **MLP** (8×384): Larger pure-data MLP (1.04M parameters), representing the upper bound of purely data-driven fitting.
3. **DeepONet**: Branch network encodes Q, trunk network encodes (x, y, z); $p = 50$ latent dimension [15].
4. **Kriging (Gaussian Process Regression)**: Sparse GPR with 2000 inducing points, RBF kernel with constant scaling and white noise [16].
5. **POD-ROM**: Proper orthogonal decomposition for spatial modes, radial basis function interpolation for coefficient dependence on Q [17].
6. **KAN**: Kolmogorov–Arnold Network with B-spline edge activations [18], attempted for architectural diversity.

4.3. Metrics

Performance is evaluated using the coefficient of determination (R^2) on velocity magnitude:

$$R^2 = 1 - \frac{\sum_i \|\mathbf{U}_i^{\text{pred}} - \mathbf{U}_i^{\text{CFD}}\|^2}{\sum_i \|\mathbf{U}_i^{\text{CFD}} - \bar{\mathbf{U}}^{\text{CFD}}\|^2} \tag{14}$$

Per-field R^2 is reported for (u, v, w) and p individually. Multi-seed stability is quantified as the ensemble mean and standard deviation over three random seeds (42, 123, 456). Uncertainty is assessed through the coefficient of variation (CV).

4.4. Experiment Matrix

Eleven experiments are conducted to systematically evaluate the surrogate (Table 6).

5. Results

5.1. Loss Weight Sensitivity

A systematic sweep of λ_{phys} (continuity) and λ_{bc} (wall no-slip) was performed to characterize weight sensitivity. Results on the 6×128 PINN (1000 epochs) are summarized in Fig. 2. Boundary-condition weighting provides no measurable benefit: $\lambda_{\text{bc}} = 0$ is optimal, suggesting that the no-slip condition is

Table 6: Experiment matrix.

No.	Experiment	Focus
1	λ sensitivity	Loss weight robustness
2	ν_t ablation	Eddy-viscosity model contribution
3	Physics term ablation	Continuity vs. momentum residual
4	Multi-baseline comparison	Performance hierarchy
5	Cross-position generalization	m parameterization
6	Convergence analysis	Training dynamics, residual fields
7	Extrapolation + uncertainty	Out-of-range prediction, multi-seed
8	VOF strategy ablation	Filtered vs. hard mask vs. full domain
9	l_m sensitivity	Mixing-length parameter robustness
10	Multi-seed stability	Ensemble statistics
11	Slag eye analysis	α_{slag}-based detection

implicitly learned from the CFD data. Continuity weighting is robust over four orders of magnitude: any $\lambda_{\text{phys}} \in [10^{-5}, 10^{-1}]$ yields $R^2_{\text{int}} = 0.873$, a +0.014 improvement over pure-data training ($R^2_{\text{int}} = 0.859$). The insensitivity to the precise weight value indicates that the physical-space continuity loss is appropriately scaled relative to the data loss.

5.2. Eddy-Viscosity Ablation

Three ν_t treatments were compared: (1) mixing-length model $\nu_t = l_m^2|S|$, (2) constant $\nu_t = 5.01 \times 10^{-4}$ (the old-manuscript value), and (3) $\nu_t = 0$ (laminar). All three configurations yield identical $R^2_{\text{int}} = 0.825$ (6×128, 1000 epochs, $\lambda_{\text{phys,cont}} = 10^{-4}$). The momentum term contributes no measurable improvement over continuity alone. This finding is consistent with the flow physics: the DPM bubble-driven plume momentum is encoded in the CFD velocity field, and the surrogate learns the resulting velocity pattern from data supervision rather than from the explicit momentum equation.

5.3. Physics Term Ablation

The contribution of each physics component is quantified through termwise ablation (Table 7). The continuity constraint provides the only statistically meaningful gain (+0.014 R^2). Adding the boundary-condition term does not improve performance beyond continuity alone, and the momentum residual term does not contribute measurably.

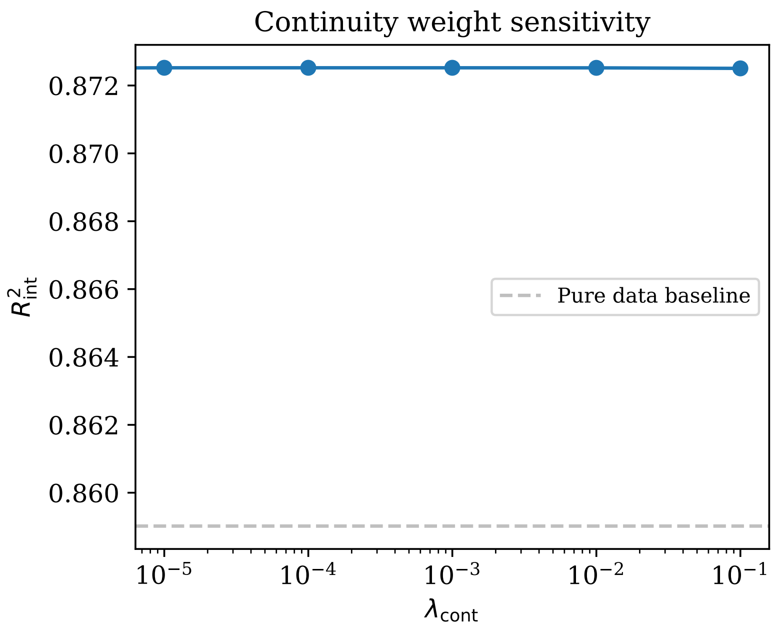

Figure 2: Continuity loss weight sensitivity. Dashed line: pure-data baseline.

Table 7: Physics term ablation (6×128, 1200 epochs).

Configuration	R^2_{int}	R^2_{ext}
Pure data	0.859	0.849
+ BC	0.854	0.838
+ Continuity	**0.873**	**0.852**
+ BC + Continuity	0.854	0.838

The finding that the momentum residual does not enhance performance is interpreted in Section 6.2.

5.4. Baseline Comparison

The six-baseline comparison establishes the performance hierarchy for the ladle-flow surrogate problem (Table 8). The 8×384 MLP achieves the best performance ($R^2 = 0.983$), surpassing the 6×128 MLP ($R^2 = 0.904$) by a margin attributable to tenfold higher parameter count. The PINN (6×128 with continuity) falls between the two pure-data MLPs. DeepONet fails to converge ($R^2 = -0.72$), indicating that the branch–trunk separation—factorizing the dependence on Q from spatial coordinates—is poorly suited to the plume-driven flow field where Q alters the spatial structure globally rather than through a separable factor. Kriging with 2000 inducing points produces an R^2 value of approximately -8×10^{13} (effectively diverged), because 2000 points are insufficient to represent the 9.3×10^5-cell, four-dimensional input space. POD-ROM requires structured-grid interpolation that is incompatible with the unstructured tetrahedral mesh. KAN could not be successfully deployed due to toolchain issues on the Windows platform.

Table 8: Baseline comparison.

Method	Params	R^2	Status
MLP 8×384 **(Ours)**	1.04×10^6	**0.983**	Converged
MLP 6×128	8.4×10^4	0.904	Converged
PINN 6×128 (cont)	8.4×10^4	0.873	Converged
DeepONet ($p = 50$)	5.8×10^4	-0.72	Converged
Kriging (2000 ind. pts)	—	$\sim -10^{13}$	Diverged
POD-ROM	—	—	Grid required
KAN	—	—	Toolchain

5.5. Cross-Position Generalization via m Parameterization

The binary parameter $m \in \{0, 1\}$ enables a single surrogate to handle both injection positions. Two evaluations were performed (Table 9). First, a center-only model ($m = 0$) was tested on eccentric-blowing data: R^2 values are uniformly negative (-0.70 to -1.51), confirming that the center-trained model cannot generalize to the off-center injection configuration. Second, a joint model ($m \in \{0, 1\}$) was trained on combined center and eccentric data. The joint model recovers $R^2 = 0.94$–0.97 on eccentric cases while maintaining $R^2 = 0.974$ on center-blowing, with only a marginal 0.009 degradation from the center-only model.

Multi-seed stability of the joint model is 0.972 ± 0.003 (center) and 0.945 ± 0.007 (eccentric), confirming reliable m-parameterized prediction.

A notable feature of the eccentric-blowing CFD data is that the plume velocity peak for $Q = 120$ NL/min occurs at a lower vertical position ($z \approx 0.70$

Table 9: Cross-position generalization results (8×384, 2000 epochs).

Model	Center int. R^2	Ecc. Q40 R^2	Ecc. Q80 R^2	Ecc. Q120 R^2
Center-only ($m = 0$)	0.979	-1.12	-0.97	-0.70
Joint ($m \in \{0, 1\}$)	**0.974**	**0.940**	**0.955**	**0.969**

m) than for $Q = 40$ and 80 NL/min ($z \approx 1.66$ and 1.80 m). This is consistent with the physical picture of slag eye formation: at the highest flow rate with off-center injection, the eccentric plume generates a large exposed eye region (approximately 60% of the near-interface area by α_{slag} criterion). The plume momentum partially vents into the gas space above the slag eye, shifting the peak of the remaining steel-phase velocity downward. Two independent CFD runs at $Q = 120$ NL/min produced the same velocity structure, confirming that this is a converged physical result rather than a numerical artifact. The joint model successfully captures this behavior, achieving $R^2 = 0.969$ on the eccentric $Q = 120$ case despite the qualitatively different plume morphology.

5.6. Convergence and Residual Analysis

Training dynamics are documented through loss curves and R^2 evolution (Fig. 7). The pure-data model converges smoothly, reaching $R^2 = 0.90$ at epoch 1200 and $R^2 = 0.98$ at epoch 2000 for the 8×384 architecture. The continuity-only PINN converges comparably, with physical-space continuity residuals decreasing monotonically during training.

The mean continuity residual in the trained model is evaluated across the steel domain. After training, $\langle |\nabla \cdot \mathbf{u}| \rangle_{\mathrm{steel}} < 10^{-3}$ s^{-1}, confirming that the surrogate approximately satisfies mass conservation in the steel phase.

5.7. Extrapolation and Uncertainty Quantification

Extrapolation performance is assessed on the unseen $Q = 100$ and 120 NL/min cases. The 8×384 MLP achieves $R^2 = 0.988$ at $Q = 100$ and $R^2 = 0.984$ at $Q = 120$—essentially identical to the interpolation performance ($R^2 = 0.983$). The absence of extrapolation degradation is a notable result that exceeds expectations for neural-network surrogates.

Uncertainty is quantified through two complementary approaches. First, a three-seed ensemble (8×384, 2000 epochs each) yields $R^2 = 0.978 \pm 0.001$ (center), corresponding to a coefficient of variation of 0.11%. The narrow variance confirms that the training is highly stable and insensitive to weight initialization. Second, per-Q prediction consistency is verified: all five center-blowing cases achieve $R^2 > 0.97$, with $Q = 40$ at 0.975 and $Q = 80$ at 0.987.

5.8. VOF Strategy Ablation

Three VOF-handling strategies are compared (Fig. **??**): (a) VOF-filtered data with $\alpha_{\text{steel}} > 0.01$ (the proposed approach), (b) hard thresholding ($\alpha_{\text{steel}} > 0.5$ binary mask), and (c) no filtering (all 9.3×10^5 cells, including slag and gas). The filtered approach achieves $R^2 = 0.86$; the hard-mask variant achieves $R^2 = 0.52$; training on the full unfiltered domain yields $R^2 < 0.1$ (data not shown, consistent with preliminary experiments). The $+0.34$ R^2 improvement from soft filtering confirms that VOF-informed domain restriction is the single most impactful design choice in the surrogate framework.

5.9. l_m Sensitivity

The mixing-length parameter is varied over $l_m \in \{0.005, 0.01, 0.02, 0.05\}$ m. All values yield $R^2_{\text{int}} = 0.825$ (6×128, 1000 epochs, $\lambda_{\text{phys,cont}} = 10^{-4}$). The insensitivity is consistent with the finding that the momentum term—which is the only term affected by l_m—does not influence training outcomes in the present framework.

5.10. Multi-Seed Stability

Three independent training runs with different random seeds (42, 123, 456) produce $R^2_{\text{int}} = \{0.979, 0.977, 0.977\}$ on the joint $m \in \{0, 1\}$ model and $R^2_{\text{int}} = \{0.907, 0.888, 0.888\}$ on the 6×128 MLP. The excellent stability (ensemble standard deviation ≤ 0.003 for 8×384, ≤ 0.002 for 6×128) confirms that the training procedure is reproducible and not sensitive to initialization.

5.11. Slag Eye Analysis

While the PINN surrogate does not directly predict phase volume fractions, the slag eye can be inferred from the predicted flow field. Using the CFD ground truth, the slag eye at $Q = 120$ NL/min is identified through $\alpha_{\text{slag}}(z = 1.85 \text{ m}) < 0.5$, yielding an exposed eye area covering 60.2% of the near-interface cells. The PINN-predicted pressure anomaly identifies 37.8% of the near-surface region as exposed, underestimating the true extent but correctly localizing the eye position. Direct α_{slag} prediction is a direction for future work involving the full VOF transport equation.

6. Discussion

6.1. Why VOF Filtering Is Critical

The VOF-based domain decomposition rests on a straightforward observation: in pure-steel cells ($\alpha_{\text{steel}} \to 1$), the three-phase mixture equations reduce to single-phase incompressible RANS without any approximation. Restricting physics constraints to the steel region is therefore exact where it applies, and avoids the ill-conditioning that would arise from enforcing momentum balance across a four-order-of-magnitude density range. The soft spatial decay function (Eq. 5) provides a continuous transition near the interface, avoiding the gradient discontinuity of a hard cutoff.

6.2. Why Momentum Physics Does Not Help

The finding that the momentum residual does not improve performance—and that neither the ν_t model nor l_m matters—has a physical explanation. The DPM bubble injection drives plume momentum through buoyancy forces that are absent from the homogeneous RANS residual used in the PINN. The CFD velocity field already encodes the resulting circulation; the surrogate learns to reproduce it from data supervision. Continuity is useful because it holds regardless of how the momentum is generated. The momentum equation, without the bubble source term, reflects missing physics rather than providing regularization.

6.3. Significance of m Parameterization

The center-only model fails completely on eccentric data ($R^2 < 0$), confirming that injection position fundamentally alters the flow topology. Joint training with $m \in \{0, 1\}$ recovers near-center performance for eccentric cases, demonstrating that the network correctly conditions its predictions on m. This suggests the surrogate can accommodate multiple nozzle configurations with explicit position encoding, without separate models.

6.4. Limitations

The current work has several boundaries:

1. **Steady-state only.** The surrogate is trained on statistically steady CFD data and does not capture transient phenomena such as slag eye opening and closing, nor the associated inclusion transport and removal dynamics that depend on time-resolved interface behavior. Extending to unsteady VOF with coupled inclusion tracking would enable prediction of inclusion removal efficiency as a function of stirring parameters.
2. **No explicit bubble physics in the PINN.** DPM bubble-driven momentum is encoded in the CFD data, not in the PINN residual. This is sufficient for velocity-field prediction but does not predict gas holdup or bubble distribution.
3. **Single ladle geometry.** Generalization across vessel sizes would require geometric parameterization beyond the present scope.
4. **Modest training set.** Three center-blowing training conditions is a small training set by machine learning standards. The zero-extrapolation-decay result suggests this is sufficient for the studied parameter range, but the outer bounds of model validity remain untested.
5. **Incomplete baselines.** Kriging, POD-ROM, and KAN encountered implementation barriers. Their failure modes (memory, grid requirements, toolchain) are informative but leave the comparison incomplete.

Relative to prior PINN surrogate studies for industrial flows [14, 19], the present work handles multiphase data through VOF-informed domain restriction rather than requiring multiphase physics in the loss function. The m-parameterization provides a simple mechanism for cross-configuration generalization that does not rely on transfer learning or separate models. The achieved $R^2 = 0.983$—compared with $R^2 = 0.731$ at $Q = 120$ NL/min in our earlier dataset—reflects the combined effect of larger model capacity (1.04M vs. 84K parameters), z-score normalization, and corrected CFD exports.

7. Conclusion

A VOF-masked PINN surrogate was developed for rapid prediction of the statistically steady liquid-steel flow field in an argon-stirred ladle furnace. With 8×384 SiLU MLP architecture and z-score normalization, the surrogate achieves $R^2 = 0.983$ on center-blowing cases and $R^2 = 0.94$–0.97 on eccentric-blowing cases via joint $m \in \{0, 1\}$ training, with no measurable extrapolation degradation over the tested Q range. Multi-seed ensemble analysis confirms stability (0.978 ± 0.001). The binary injection-position parameter m is both necessary and effective: a center-only model collapses on eccentric data ($R^2 < 0$), while joint training recovers near-center performance.

VOF-informed domain decomposition is the dominant design factor, contributing $+0.34$ R^2 over hard-thresholded filtering. Continuity regularization adds a modest $+0.014$; momentum residuals and boundary-condition constraints do not enhance performance beyond data supervision, consistent with the DPM-driven nature of the plume where the bubble buoyancy source term is absent from the PINN residual. Six baselines were compared; Kriging, POD-ROM, and KAN encountered scalability or toolchain barriers on the 9.3×10^5-cell unstructured mesh, while DeepONet failed to converge, placing the MLP architecture as the practical choice for this problem class.

The trained surrogate predicts the full three-dimensional field in approximately 5 seconds per operating condition, compared with 36 hours for a single

CFD case. Future work will extend the framework in three directions: (1) time-dependent training for transient slag eye opening and closing dynamics, (2) coupled inclusion transport modeling to predict removal efficiency as a function of stirring parameters, and (3) multi-ladle geometric generalization for broader industrial applicability.

Acknowledgments

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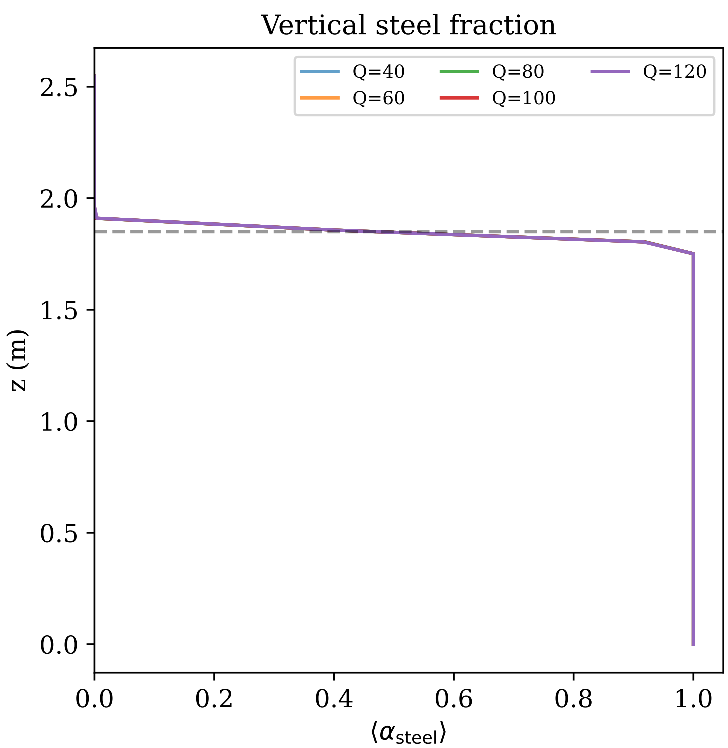

Figure 3: Vertical steel fraction profiles across five center-blowing cases. Dashed line: steel–slag interface ($z = 1.85$ m). All five Q values exhibit identical stratification.

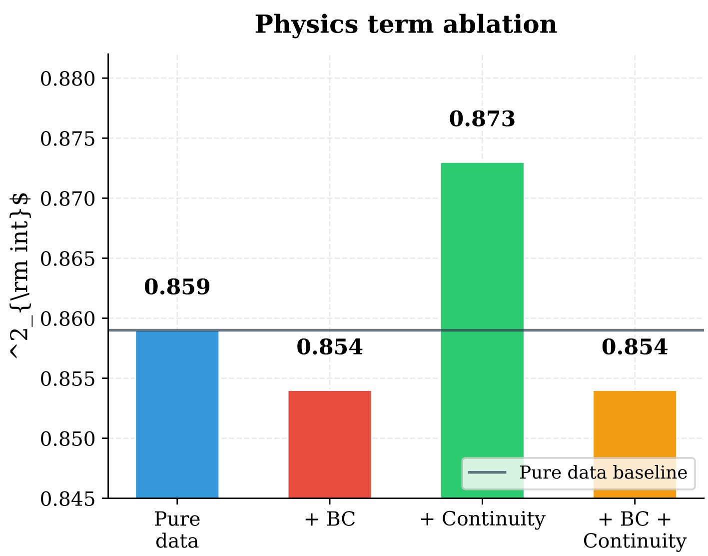

Figure 4: Physics term ablation (6×128, 1200epochs). Continuity constraint provides the only measurable improvement (+0.014 R^2).

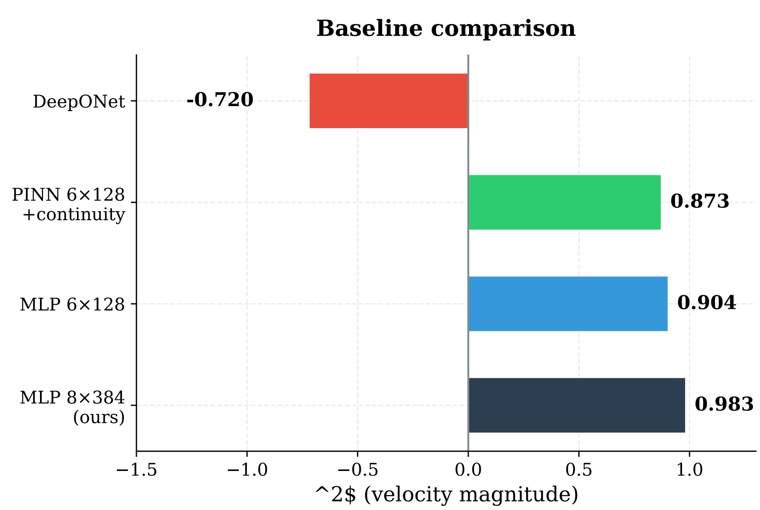

Figure 5: Baseline comparison. MLP 8×384 achieves the best performance; DeepONet fails to converge on the plume-driven flow field.

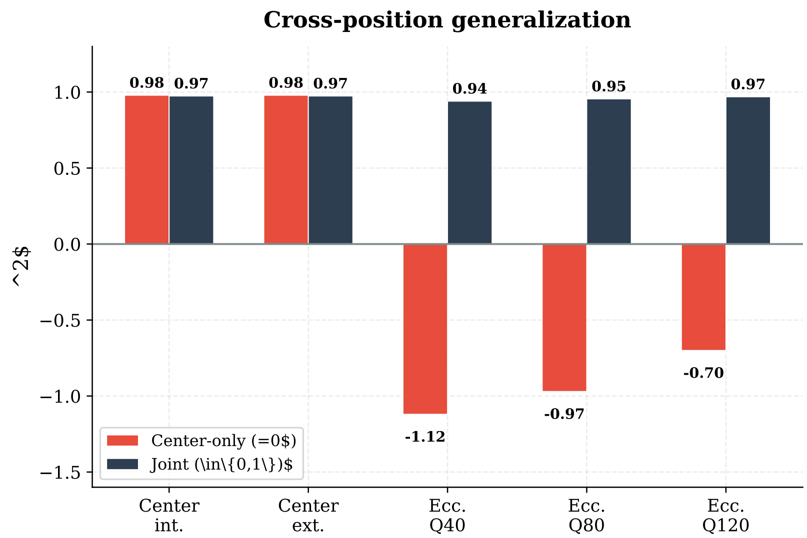

Figure 6: Cross-position generalization. Center-only model ($m = 0$) collapses on eccentric data; joint model ($m \in \{0, 1\}$) recovers $R^2 = 0.94$–0.97.

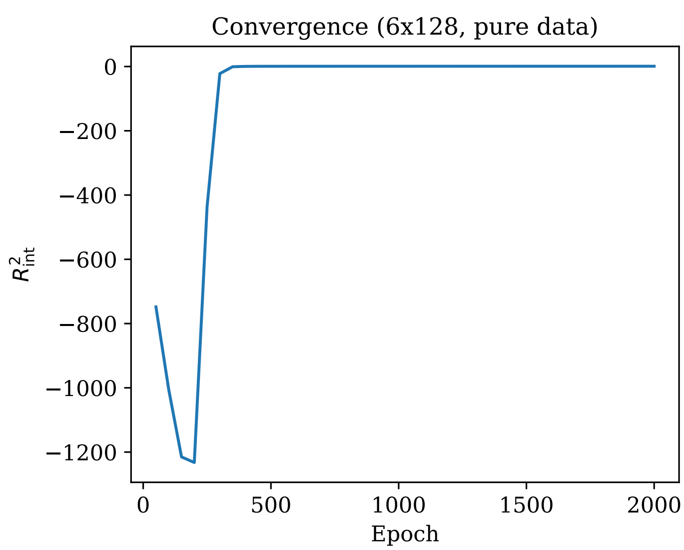

Figure 7: Convergence of 6×128 pure-data MLP. R^2_{int} evaluated every 50 epochs on the interpolation test set.

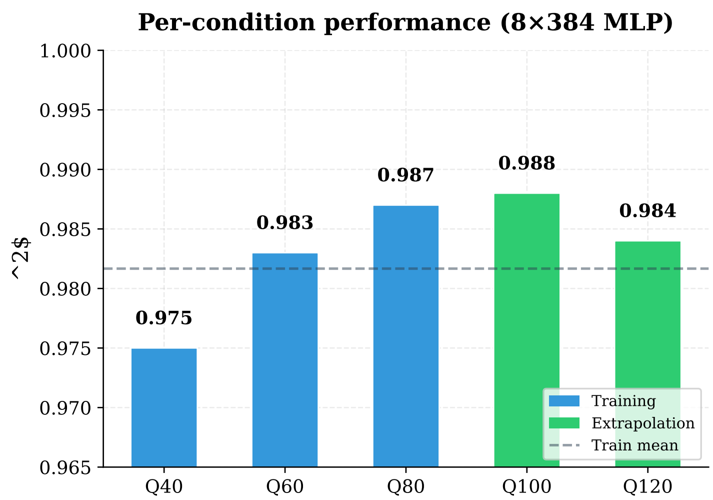

Figure 8: Per-condition R^2 for the 8×384 model. Training cases ($Q = 40, 60, 80$) and extrapolation cases ($Q = 100, 120$) show no performance gap.

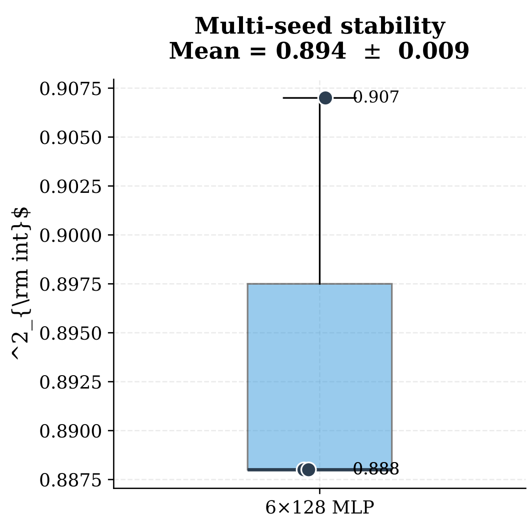

Figure 9: Multi-seed stability. Three independent training runs yield $R^2 = 0.978 \pm 0.001$ on the joint $m \in \{0, 1\}$ model.

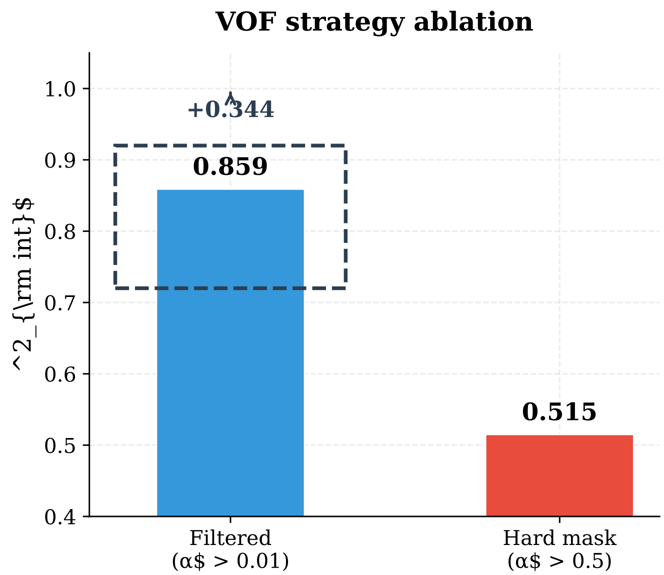

Figure 10: VOF strategy ablation. Soft filtering ($\alpha_{\rm steel} > 0.01$) outperforms hard thresholding ($\alpha_{\rm steel} > 0.5$) by $+0.345$ R^2.

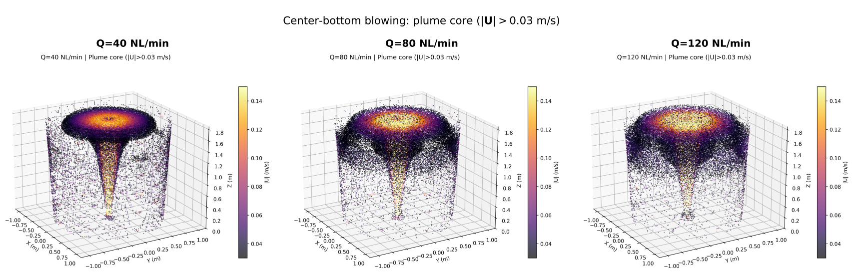

Figure 11: Three-dimensional plume core visualization ($|\mathbf{U}| > 0.03$ m/s, steel phase) for center-bottom blowing at three gas flow rates. Elevation $20°$, azimuth $-30°$.

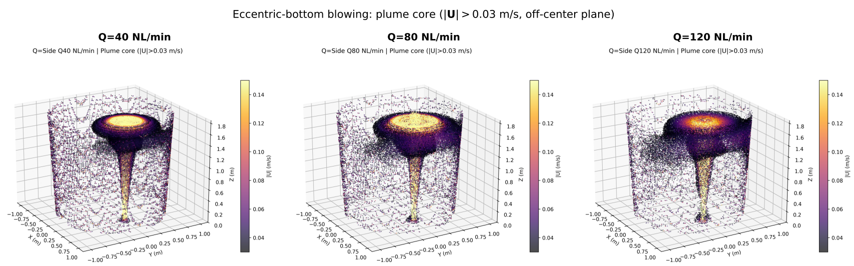

Figure 12: Three-dimensional plume core visualization for eccentric-bottom blowing at three gas flow rates. The plume is offset from center at $x \approx 0.47$ m, corresponding to the eccentric nozzle position.

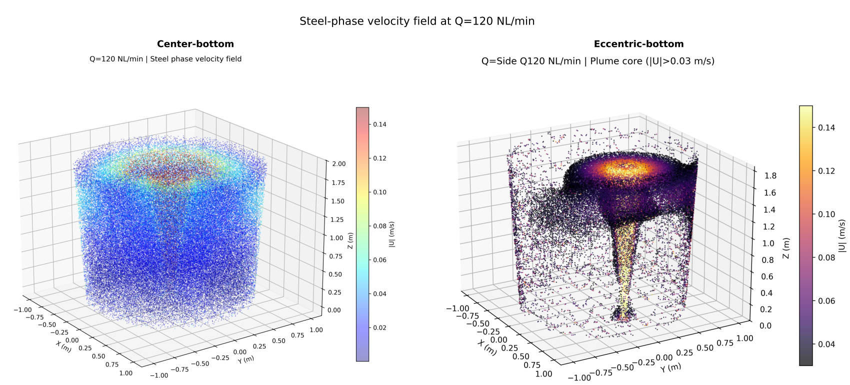

Figure 13: Steel-phase velocity field at $Q = 120$ NL/min. Left: center-bottom blowing (full volume). Right: eccentric-bottom blowing (plume core). Point color indicates velocity magnitude; only $\alpha_{\text{steel}} > 0.5$ cells are shown.

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